MILITARY SPECIFICATION

DECODER GROUP, AN/UPA-59(V) IFF MK XII (SIF), WITH ACTIVE READOUT AND ASSOCIATED MODIFICATION TO INDICATOR (PPI)

This amendment forms a part of Military Specification MIL-D-28736B(EC), dated 20 December 1971.

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* 3.2.3.3: Add the following to the end of the paragraph: "A positive locking device shall be furnished to hold the chassis while extended for maintenance. This locking device shall be safe and effective for horizontally mounted decoders, as well as those mounted up to and including ±90 degrees from horizontal.

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3.4.10, first sentence: Delete and substitute: "The combined weight of the decoder and active readout unit shall not exceed 32 pounds". Delete last sentence.

3.4.13: Delete and substitute the following:

"3.4.1.3 Mounting provisions for brackets and mounting plates. Mounting brackets and mounting plates shall be furnished by the contractor. Proposed mounting provisions shall be approved by the command or agency concerned prior to use (see 6.3).".

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3.4.15.10.4, Delete and substitute:

"3.4.15.10.4 Identification tag decode tolerance. Mode 1, 2, 3/A, 4 and C tag decoders shall produce a tag decode when the proper pulse configuration (nominal pulse width) is received over ± 0.35 µsecond of the ideal spacing. For ± 0.65 µsecond from ideal spacing, no decode shall be produced.".

* Add the following new paragraph:
"3.4.15.10.5 Decode reset. In the absence of reset pulses, it shall be possible to reset the decode from the P1 pulse of the identification tags. This format of operation shall be selected by a solderless jumper link or one circuit card within the basic decoder.

3.4.15.12: Delete "± 0.55 µsecond" and substitute "± 0.65 µseconds.".

3.4.15.15, Delete all but the first sentence and add "(see 3.4.27.)"

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous amendment.

FSC 5895

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3.4.18: Add as first sentence:

"Controls, indicators, and panel layouts shall be submitted to the Command or Agency concerned for approval (see 6.3).".

3.4.18(a)(8): Delete and substitute:

"3.4.18(a)(8) A continuously variable control, variable from 1 to 20 nautical miles, used to increase and decrease the range (length) of the target sector gate. The control may be accomplished in a maximum of two steps by use of a solderless link to select the operating range. The two ranges to be selected shall be 1-10 nautical miles and 2-20 nautical miles. The unit shall be shipped with the link in the 2 to 20 nautical mile range position.".

* 3.4.18(a)(11): Delete and substitute:
"3.4.18(a)(11) Bracket on-off-mix switch; a three position switch to select bracket decode out put.

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3.4.18(b)(16): Delete and substitute:

"3.4.18(b)(16) Spare video connector: quantity one, wired as an input to mixer amplifier.".

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3.4.23: Add the following:

"For code video trains with pulse width between 0.30 microseconds and 0.45 microseconds the output code pulse width shall be 0.45 plus or minus 0.10 microseconds. For code video trains with pulse width greater than 0.45 microseconds the code video output pulse width shall be equal to the input pulse width with a tolerance of plus or minus 0.10 microseconds.".

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3.4.27, Delete and substitute:

"3.4.27 Mode enable outputs and controls. Provisions shall be made to provide six separate mode enable outputs for challenge control of Modes 1, 2, 3/A, C, 4, and Mode 4 override. The mode enable outputs shall have the following enable control provisions:

(a) The mode enable outputs for Modes 1, 2, 3/A, C, and 4 shall have separate enable controls with any combination of these controls selectable. The selections shall be solderless links.

(b) The Mode 4 and Mode 4 override outputs shall be simultaneously enabled when the Mode 4 override control is placed in the momentary position.

(c) When the Mode 4 override control is in the sector position, the Mode 4 and Mode 4 override outputs shall be simultaneously enabled by and for the duration of the azimuth gate.

(d) Using the mode enable outputs for Modes 1, 3/A and 4, provide by means of solderless links three mode enable outputs for independent mode enable control of the six modes. The three wire output mode enable control circuits shall consist of diodes connected as shown in Figure 5 with the combined Modes 1 and 2 on the Mode 1 output line, combined Modes 3/A and C on the Mode 3/A output line, and combined Mode 4 and Mode 4 override on the Mode 4 output line.".

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3.4.29: Delete and substitute the following:

* "3.4.29 Self-test. A self-test feature shall be provided by front panel selector switch(s) for the decoder group. Insertion, decoding, and display of the front panel selected test code configuration on a one code at a time basis shall be considered adequate to test the passive, and active decode function and the display. This test feature shall also test the selected attitude layer and emergency alarm function, as described herein. Front panel selection shall also be provided for conversion of the inserted code to attitude in feet for the active readout display test of the Mode C conversion. This test switch shall be used for front panel test control and shall have three positions designated SIF, OFF and Mode C. The design of this self-test feature shall be approved by the command or agency concerned prior to use (see 6.3). When the self-test feature is selected, it shall not be necessary to set the read gate switch to ON in order to perform active testing."

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* 3.6 Function. The decoder group shall utilize, as a general guide for basic design, the functional block diagrams of figures 2 and 3 which are supplied for information only and are not represented as complete in all particulars of the requirements herein. Definitions used in this specification shall be as detailed in MIL-STD-751 (see 6.4). The decoder group shall process SIF coded replies, develop passive and bracket decode outputs, and display the active readout of reply codes. The active readout feature will serve for rapid code identification of an unknown IFF target. There shall be the capability to display simultaneously all SIF targets by means of bracket decode, and selected targets with passive decode. The passive-decode targets shall be stretched with respect to the bracket targets. When both selected target and brackets are desired, the selected target shall be stretched in pulse width for presentation. When either bracket or selected targets are used separately, normal decode pulse width shall be presented. When all brackets are selected, brackets shall be displayed for all video replies reaching the decoder, not merely those associated with modes enabled at the decoder itself. The degarble circuitry shall reduce the probability of false target display or false readout to a minimum. No pulse regeneration or trigger circuitry shall be enabled until a decision has been made by the decoder that a valid reply pulse is present. A valid pulse is one which has a pulse width of 0.30 seconds or more. Mode 4 continuous will be challenged in readout gate azimuth only. Mode 4 selected by the Mode switch will be similar to other interlaced modes. In this mode of operation selected SIF codes would be stretched, Mode 4 and bracket decode, would not be stretched.

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* 3.6.1.3(b): Delete and substitute:
"(b) Bracket video on all mode bracket video.

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3.6.1.5: Delete the last sentence and substitute the following:

"The degarbling feature shall be operative during active readout except when passive decode is connected to the early decode line; however, a simplified technique such as card reversal shall be provided to allow enabling or disabling the degarbling during passive type of operation.".

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* 3.6.1.6.2: Add the following new subparagraph:

"(c) A positive or negative dc bias of up to 2.5 volts may be present on this input. This bias shall not interfere with normal operation of the decoder, nor shall it be passed for display."

* 3.6.1.6.3: Add the following new subparagraph:

"(d) Accept unseparated video with or without resets on one cable and mode identification tags on a separate cable."

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* 3.6.1.6.4.1: Add the following new subparagraph:

"(c) Range strobe output of a PPI is frequently counted down two or more times with respect to IFF sweeps. In these cases, the decoder shall automatically continue the gating function at the range determined by the most recent range strobe, and shall maintain this position until new range strobe information is provided from the PPI. This automatic range strobe function shall require no link settings or adjustments for operation."

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3.6.1.8: Delete and substitute:

"3.6.1.8 Individual pulse shift. When any nominal pulse (0.45 µsecond width) of the correct configuration is shifted from its nominal position by ± 0.15 µseconds, the decoder shall provide an output for at least 99 percent of the replies received. For a pulse shift of ± 0.18 µseconds, the decoder shall provide an output for at least 90 percent of the replies received. For a pulse shift of ± 0.40 µseconds the decoder shall provide outputs for less than 10 percent of the replies received. When the input pulse width is decreased to 0.30 µseconds, the limits shall be the same as those specified above for 0.45 µsecond pulses. When the pulse width is increased to 1.0 µseconds, the reject limit shall increase from ± 0.40 µseconds to ± 1.0 µseconds, The accept limits for 1.0 µseconds pulses shall be the same as that specified for 0.45 µsecond pulses.".

3.6.1.9: Delete and substitute:

"3.6.1.9 Extraneous pulses. For extraneous pulses occurring within ± 0.15 µseconds of a nominal space position in a correctly decoded reply train, the decoder shall provide outputs for less than 1 percent of the reply trains. For extraneous pulses which occur at ± 0.18 µseconds from a nominal space position, the decoder shall provide outputs for less than 10 percent of the reply trains. For extraneous pulses which occur greater than ± 0.40 µseconds from a space position the decoder shall provide outputs for at least 90 percent of the reply trains.".

Page 25

3.6.1.10, fourth line: Delete "-3 dB change in the decode amplitude" and substitute the following:

"10 percent decrease in the decode pulse repetition frequency.".

Page 26

* 3.6.1.15: Delete and substitute:

"3.6.1.15 When sector challenge is used, targets located within the azimuth width of the sector gate shall be actively displayed. With the passive challenge switch on, continuous challenge shall be provided; and the IFF target display shall not be affected by the target sector gate. In either switch position, active readout of target replies is enabled only for time intervals equivalent to the size of the displayed gate. The active readout shall be retained on the channels until reset. A continuously variable internal adjustment shall be provided for active readout hold duration permitting reset time of a minimum of one second to a maximum of ten seconds following the active decode display. For cases in which the antenna's high rpm causes a second readout gate while the readout hold duration is still on, the following actions shall take place:

(a) Codes held and displayed on the active readout shall be extinguished.
(b) The active readout shall be reset.
(c) Active readout shall be provided for target replies within the second readout gate.
(d) Active readout of these new targets within the second readout gate shall continue for the full readout hold duration, or until a third readout gate occurs, upon which steps (a), (b), (c) and (d) shall be repeated."

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* 3.6.1.16: Add the following to the end of the paragraph:

"The sector gate shall operate with antennas running at speeds from 3 to 65 rpm. For antenna speeds within this range, the sector gate shall operate within the following tolerances:

(a) Sector gate size shall vary by no more than ± 5 degrees
(b) Sector gate position shall vary if no more than ± 3 degrees
(c) Position is defined by the superposition of sector gate leading edge with PPI cursor bearing leading edge."

Page 31

3.6.4.2.8, last sentence: Delete and substitute the following:

"No circuitry shall be used in the alarm decoding circuits which might cause the effectiveness of the alarm to be decreased.".

Page 32

3.6.4.2.8.1, first line: Delete "reduce the effectiveness of many" and substitute "minimize".

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6.3: Add the following new subparagraphs:

"(k) Mounting provisions for brackets and mounting plates (see 3.4.13).

"(l) Controls, indicators, and panel layouts (see 3.4.18).

"(m) Self test feature (see 3.4.29).".

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* Figure 4.: Add the following and delete the last vertical line on the right under the displayed video column:

	SAL	STRTCH	BRKT	DECODE OFF CODE	DISPLAYED VIDEO
" 25	OFF	OFF	ALL	OFF	Std brackets for all modes being challenged by the interrogator

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Add attached new figure.

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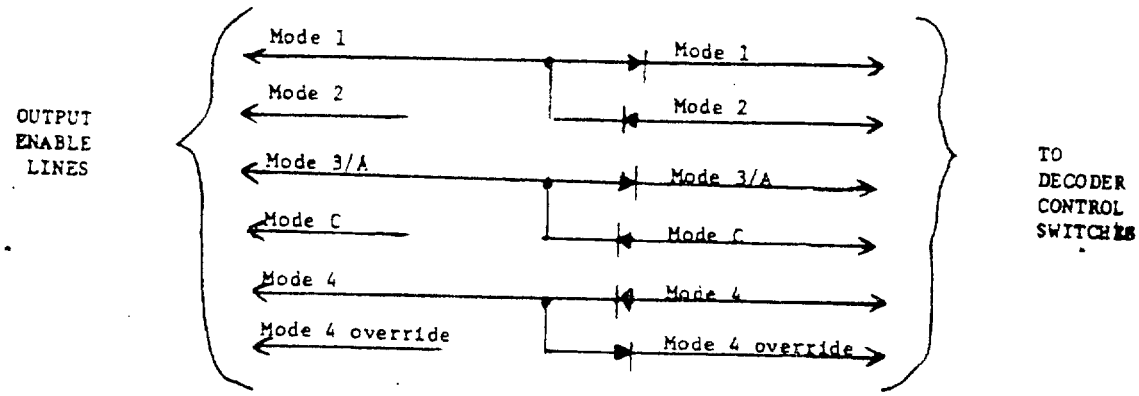


FIGURE 5. Mode enable outputs.

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